

Pihu Jain

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SUMMARY

I am a Computer Science student with a curious and practical approach to learning. I enjoy understanding how technology works and applying that knowledge to solve problems. I am comfortable learning new technologies, working both independently and in teams, and adapting to unfamiliar challenges. I value clear thinking, continuous improvement, and building a strong technical foundation through hands-on experience.

EDUCATION

Vellore Institute of Technology

CGPA: 8.64

B.Tech, Computer Science and Engineering (Cloud Computing and Automation)

Aug 2023 – May 2027

St. Joseph's Convent Hr. Sec. School, Itarsi

Class XII: 72.8% | Class X: 87.6%

2021 – 2023

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

Machine Learning & AI: Generative AI, Large Language Models(LLMs), AI API Integration, Scikit-learn, Deep Learning, NLP, Model Evaluation

Databases: PostgreSQL, MySQL, SQL Queries

Software Engineering: Data Structures and Algorithms, Object-Oriented Programming, REST APIs

Tools & Platforms: Git, GitHub, Streamlit, AWS, CI/CD

ACADEMIC PROJECTS

GapstractAI | AI-Powered Research Analysis Tool | *Python, LLM APIs, PostgreSQL, React, GitHub* | 2025 – Present
Node.js

- Developed a **Generative AI** tool using multiple **LLM APIs** (OpenAI, Claude and Gemini) to analyze research papers, generate structured summaries, track citations and identify potential research gaps.
- Implemented structured AI outputs and **schema validation** to detect missing fields before storing results in PostgreSQL, improving data consistency and reliability.

StegoX | Secure Multimedia Communication Platform | *Python, Whisper AI, AES-256, Streamlit GitHub* | 2026– Present

- Developed a Python-based application using Streamlit, integrating **AI-powered voice processing** and AES-256 encryption for secure multimedia communication.
- Implemented **image, audio and video processing** workflows for multiple media formats, and performed **testing, debugging and performance optimization** to improve application reliability.

ChokeHelm | Autonomous Well-Control System | *Python, NumPy, SciPy, Streamlit* [GitHub](#) | 2026 – Present

- Developed an **automated control system** combining target optimization and predictive control to select hourly valve settings from four noisy sensor readings, recording the reason for each adjustment.
- Evaluated performance against two classical control approaches, maintaining pressure limits where a PID controller breached them **120 times**, reducing valve movement by **29%** and settling time from 39 to 17 hours while sacrificing approximately 5% of output.

ACHIEVEMENTS AND CERTIFICATIONS

- 2× college-level hackathon winner, building and shipping working solutions under time pressure.
- Selected for the internal round of Smart India Hackathon (SIH) 2024.
- Technical Content team member, IoT Club, VIT Bhopal.
- Helped plan college events with the Ekfraseis Dramatics Society, coordinating with teams.
- Volunteered with orphaned children for a month, teaching basic subjects, organizing activities, and spending evenings mentoring and engaging with them.